signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
omega = 2 * %pi * k / N;
z = exp(%i * omega);

// Zero-pad the sequence to N points
x_padded = [x, zeros(1, N-L)];

// Compute DFT (Z-transform on the unit circle)
Xz = fft(x_padded);

// Plot magnitude spectrum
subplot(2,1,1);
plot(omega/(2*%pi), abs(Xz), "b");
xgrid();
xtitle("Magnitude Spectrum", "Normalized Frequency", "|X(z)|");

// Plot phase spectrum
subplot(2,1,2);
phase_X = atan(imag(Xz), real(Xz));
plot(omega/(2*%pi), phase_X, "r");
xgrid();
xtitle("Phase Spectrum", "Normalized Frequency", "Phase (rad)");

// Numerical Inverse Z-Transform using inverse DFT
x_rec = real(ifft(Xz));

disp("Original sequence:");
disp(x);

disp("Recovered sequence from Inverse Z-Transform:");
disp(x_rec(1:L));

```

Graphic window number 0

File Tools Edit ?

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